Optical Networks

Network Infrastructures
MSc. Cybersecurity

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SAPIENZA
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From a Logical Link to an Optical Connection

- A transport network provides connectivity between geographically separated client devices.
- Consider two routers, R_1 and R_2, located far away from each other.
- At the IP layer, the two routers may appear directly connected, with their interfaces belonging to the same IP subnet.
- However, a logical IP link requires an underlying physical transport service to carry traffic.
- This service can be provided by an optical network.



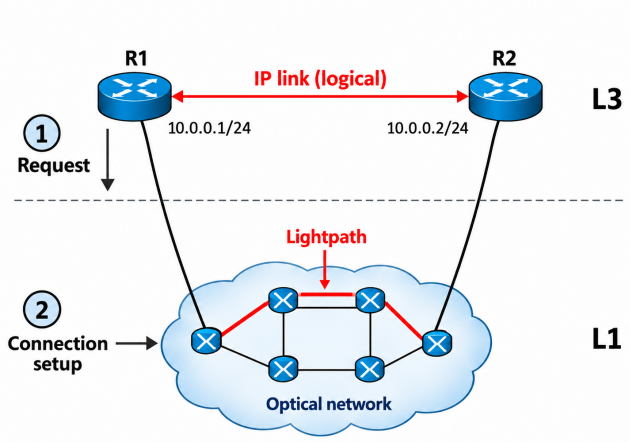
Request and Establishment of an Optical Connection

- The optical service considered here is connection-oriented.
- Before traffic can be transported, the client requests a connection between two endpoints.
- The provider must:
  - select a route,
  - reserve a channel on each traversed link,
  - configure intermediate optical devices.
- Once established, the end-to-end optical connection is called a lightpath.

Client–provider abstraction

From the routers' point of view, the service behaves as a direct link; the sequence of optical nodes remains hidden inside the provider network.

Request and Establishment of an Optical Connection





Frequency and Wavelength

- Light is an electromagnetic wave that can be described by its frequency f or wavelength λ.
- In vacuum, the two quantities are related by:

$$c = \lambda f \qquad \Longrightarrow \qquad \lambda = \frac{c}{f}$$

  where $c \simeq 3 \times 10^8$ m/s.
- Higher frequencies correspond to shorter wavelengths, and vice versa.

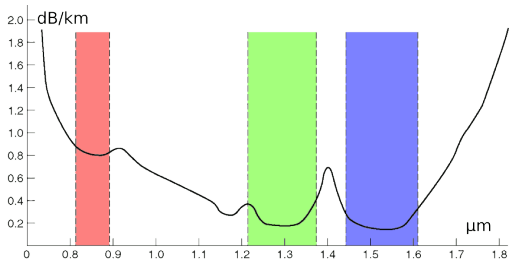
Example: Optical carrier at 1550 nm

$$f = \frac{3 \times 10^8}{1550 \times 10^{-9}} \simeq 193.5 \text{ THz}$$



Optical Transmission Windows

- Fiber attenuation depends on the wavelength of the optical signal.
- Optical communication systems exploit spectral regions characterized by low attenuation.

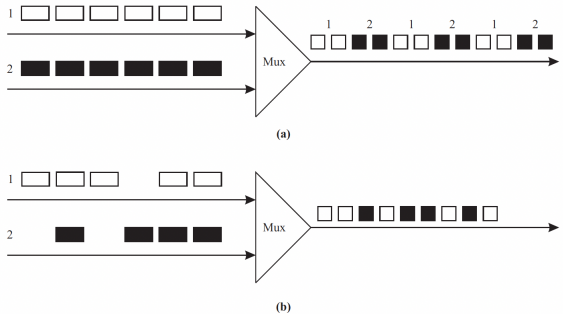




Circuit and Packet Switching

1 ON Basics

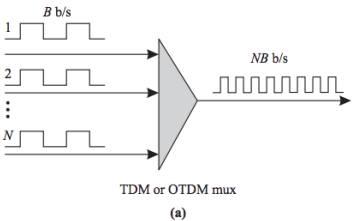
- Core devices work according to one of the following switching paradigm:
  - circuit switching: static multiplexing
  - packet switching: statistical multiplexing

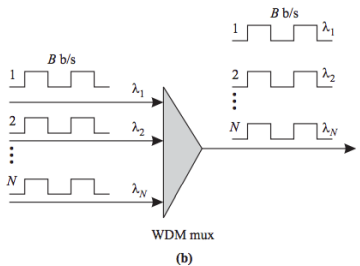


Multiplexing Techniques

1 ON Basics

- Time Division Multiplexing (TDM)
  - increase the bit rate (requires higher-speed electronics)
  - many lower-speed data streams are multiplexed into a higher-speed stream
- Wavelength Division Multiplexing (WDM)
  - transmit data simultaneously at multiple carrier wavelengths over a fiber
  - virtual fibers

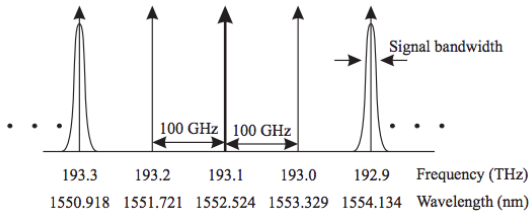






Optical Carriers and Channel Spacing

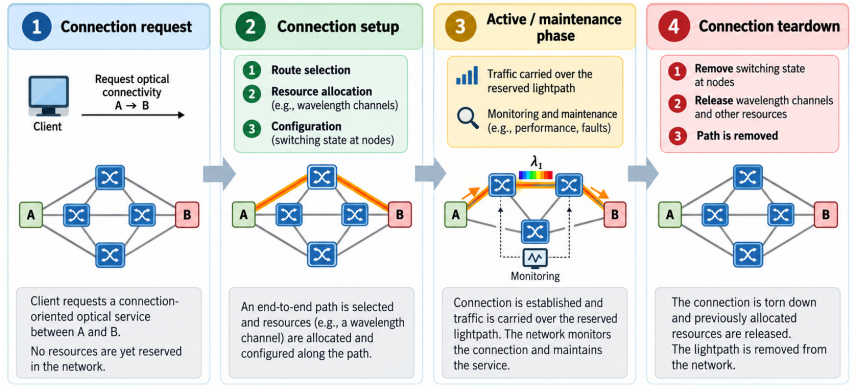
- In WDM systems, carriers are typically arranged on a regular frequency grid.



- **Channel spacing**: separation between adjacent carrier center frequencies.
- **Signal bandwidth**: spectral region occupied by the modulated signal.

Optical Connection Set-up

1 ON Basics





Optical Networks

1 ON Basics

- Optical Networks (ONs) can deliver bandwidth in a flexible manner where and when needed
- Optical fiber
  - offers much higher bandwidth than copper cables
  - is less susceptible to various kinds of electromagnetic interference and other undesirable effects
- Two generations of optical networks
  - First generation
    - optics essentially used for transmission and simply to provide capacity
    - switching and other intelligent network functions handled by electronics
  - Second generation
    - routing, switching, and intelligence in the optical layer



Second-Generation Optical Networks

- Also known as wavelength routed networks
- Main idea
  - incorporate some of the switching and routing functions into the optical part of the network
- The network provides lightpaths to its users
- Lightpaths are optical connections
  - carried end to end from a source node to a destination node
  - over a wavelength on each intermediate link
- At intermediate nodes the lightpaths are switched from one link to another link
- Lightpaths may be converted from one wavelength to another wavelength

Second-Generation Optical Networks

1 ON Basics

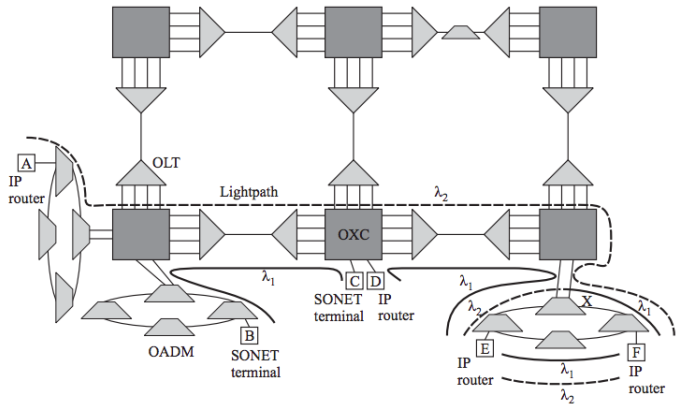




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Optical Line Terminals

2 ON Devices

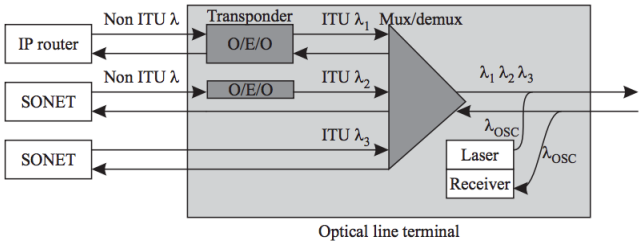
- OLTs are used at either end of a point-to-point link to multiplex and demultiplex wavelengths
- It is composed by three functional elements:
  - Transponders
  - Wavelength multiplexers
  - Optical amplifiers
- Transponder (a.k.a. optical-to-electrical-to-optical, O/E/O) adapts the signal coming in from a client of the optical network, and vice versa
  - converts the signal into a wavelength that is suited for use inside the optical network (from 1.3 μm to 1.55 μm)
  - adds OTN overhead (OPU, ODU, OTU, FEC, etc.)
  - monitors the bit error rate of the signal at the ingress and egress points in the network
- OLT also terminates an optical supervisory channel (OSC)



Optical Line Terminals

2 ON Devices

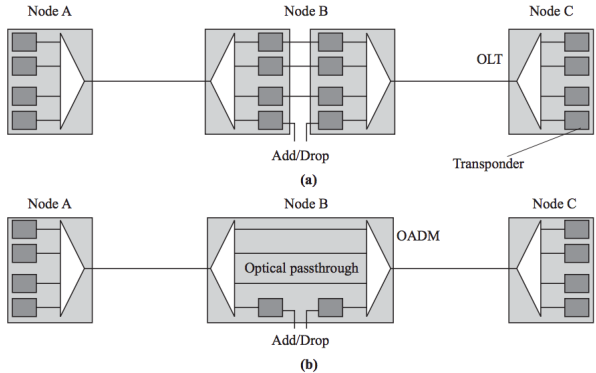
- Transponders typically constitute the bulk of the cost, footprint, and power consumption in an OLT
- Therefore, reducing the number of transponders helps minimize both the cost and the size of the equipment deployed



Optical Add/Drop Multiplexers

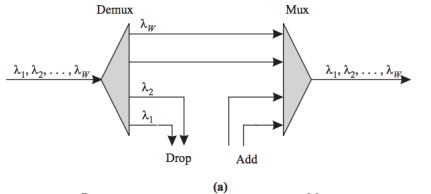
2 ON Devices

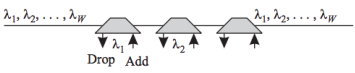
- Optical add/drop multiplexers (OADMs) provide a cost-effective means for handling passthrough traffic in both metro and long-haul networks

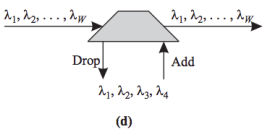


OADM Architectures

2 ON Devices



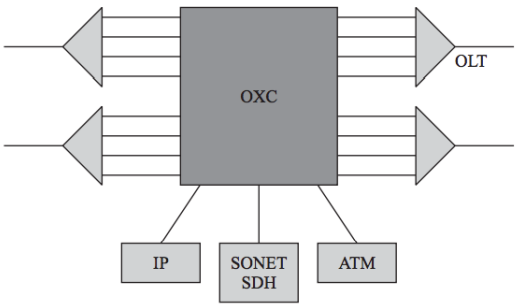




Optical Crossconnects

2 ON Devices

- OXC enables reconfigurable optical networks, where lightpaths can be set up and taken down as needed
- OXCs allow to handle complex topologies and large number of wavelengths





Optical Crossconnects

2 ON Devices

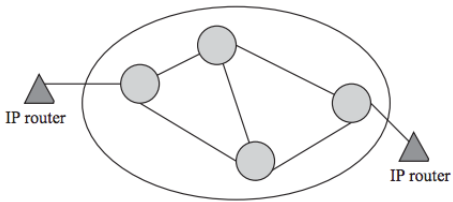
- An OXC provides several key functions in a large network:
  - service provisioning
    - provisioning of lightpaths in a large network in an automated manner (no manual patch panel connections)
    - reconfigure lightpaths to respond to traffic changes
  - protection
    - detect failures and rapidly reroute lightpaths around the failure
  - bit rate transparency
    - switch signals with arbitrary bit rates and frame formats
  - wavelength conversion
    - change the wavelength of an incoming signal before transmitting it
  - multiplexing and grooming
    - multiplexing and grooming capabilities to switch traffic internally at much finer granularities
    - this time division multiplexing has to be done in the electrical domain

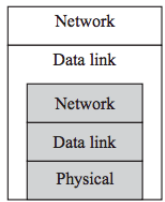


The Optical Layer

2 ON Devices

- Network architectures can be organized by means of the ISO/OSI model
- A more realistic layered model for today's networks would employ multiple protocol stacks residing one on top of the other







Transparency and All-Optical Networks

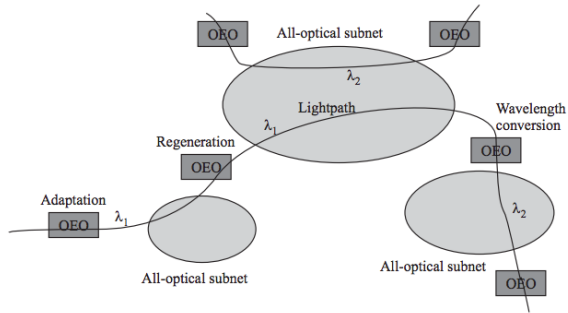
2 ON Devices

- Lightpaths are service transparent
  - once the lightpath is set up, it can accommodate different types of services
    - e.g., the telephone network had this property (a channel can be used to transfer voice, data, fax, etc.)
- Advantages:
  - data is carried from its source to its destination in optical form
  - no optical-to-electrical conversions along the way
- Hard to realize:
  - analog signals require higher SNR with respect to digital ones
- Optical networks almost always include a fair amount of electronics

Transparency and All-Optical Networks

2 ON Devices

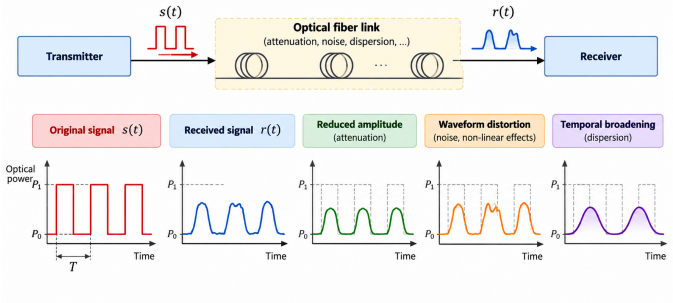
- Electronics plays a crucial role in control and management functions.
- Electronics is required:
  - **At the network edge**
    - To adapt signals entering the optical domain.
  - **In the network core**
    - For regeneration and wavelength conversion.



Signal Degradation in Optical Networks

2 ON Devices

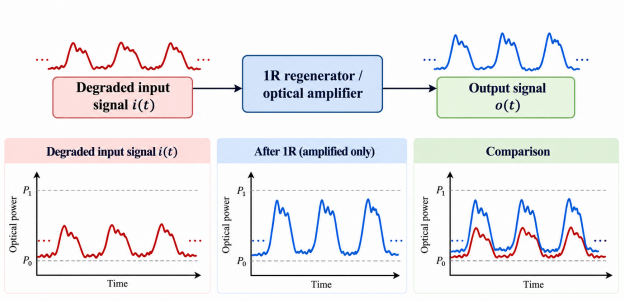
- Optical signals progressively degrade as they propagate along a lightpath.
  - **Attenuation**: reduction of optical power.
  - **Distortion**: alteration of the signal waveform.
  - **Dispersion**: temporal broadening of optical pulses.



1R Regeneration: Reamplification

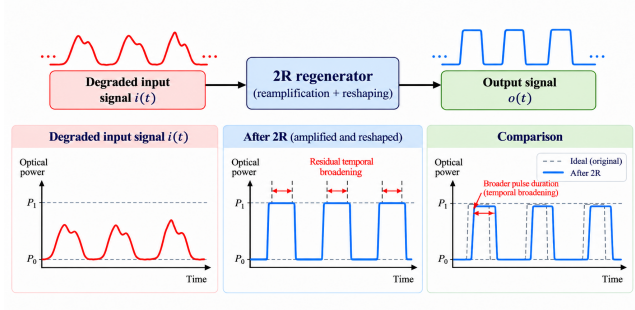
2 ON Devices

- Restores the optical power of the received signal.
- Does not correct waveform distortion or timing errors.
- **PRO**
  - Supports analog signals.
  - No O/E/O conversion required.
- **CONS**
  - Distortion and noise are not removed.



2R Regeneration: Reamplification and Reshaping

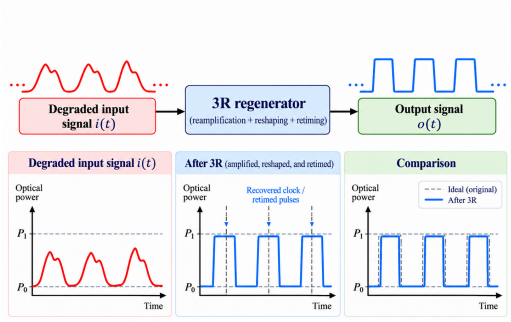
- Restores signal amplitude and pulse shape.
- Requires O/E/O conversion to reconstruct digital pulses.
- **PRO**
  - Can preserve bit-rate transparency.
- **CONS**
  - Timing errors are not corrected.
  - Jitter may accumulate across multiple regenerators.



3R Regeneration: Reamplification, Reshaping and Retiming

2 ON Devices

- Restores amplitude, pulse shape and timing.
- Recovers the clock and resynchronizes the digital signal.
- **PRO**
  - Produces a “fresh” copy of the digital signal.
- **CONS**
  - Requires knowledge of the bit rate.
  - Reduces transparency to bit rates and signal formats.



Network Evolution

2 ON Devices

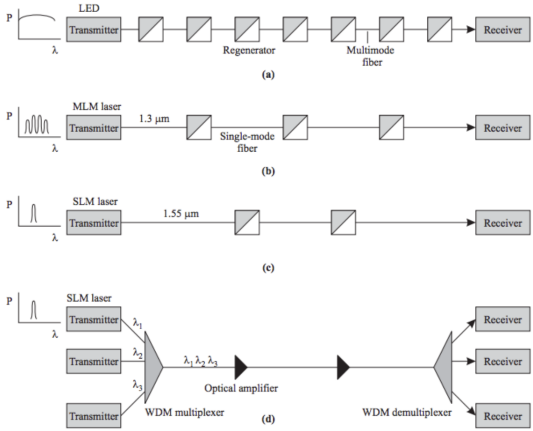




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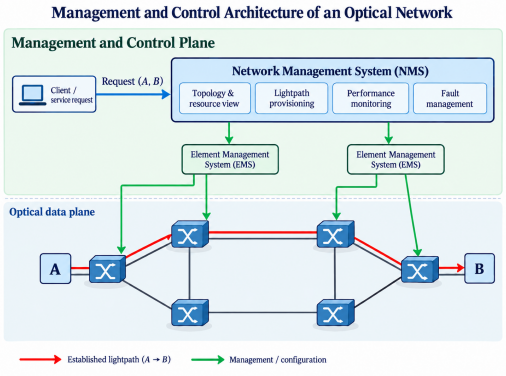
▶ Control and Management



Management Framework

3 Control and Management

- Most functions of network management are implemented in a centralized manner by a hierarchy of management systems





Optical Layer Services and Interfacing

3 Control and Management

- Client layers can specify to the optical layer the following services during lightpath setup:
  - the endpoints to interconnect
  - the amount of bandwidth that is required
  - it can specify if an adaptation function is needed at the ingress or egress point
  - the targeted Bit Error Rate (BER)
  - the level of protection against failure events
  - requirements related to jitter and maximum end to end delay
- Enabling the delivery of these services requires a control and management interface between the optical layer and the client layer
  - The simple interface used today is through the management system
  - It works fine as long as lightpaths are set up fairly infrequently and remain nailed down for long periods of time



Layers within the Optical Layer

3 Control and Management

- The optical layer performs several functions (multiplexing, switching and routing, and performance monitoring)
- In order to help delineate management functions it is useful to further subdivide the optical layer into several sublayers:
  - *Optical Channel* layer (OCh):
    - takes care of end to end routing of the lightpaths
  - *Optical Multiplex Section* layer (OMS):
    - each link between OLTs or OADMs represents an optical multiplex section carrying multiple wavelengths
  - *Optical Transmission Section* layer (OTS):
    - link between two optical amplifier stages



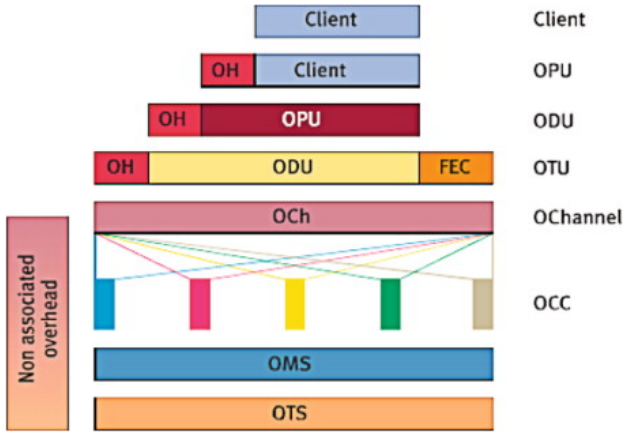
Layers within the Electronic Layer

3 Control and Management

- Optical channel transport unit (OTU)
  - delineate OTN frames
  - provide identification of the optical connection
  - monitor bit error rate (BER) performance
  - carry alarm indicators to signal failures
  - provide a communication channel between the end points of the optical connection
- Optical channel data unit (ODU)
  - as OTU, but at a higher layer
  - includes the Optical channel Payload Unit (OPU) sublayer that adapts client signals to the OTN frames

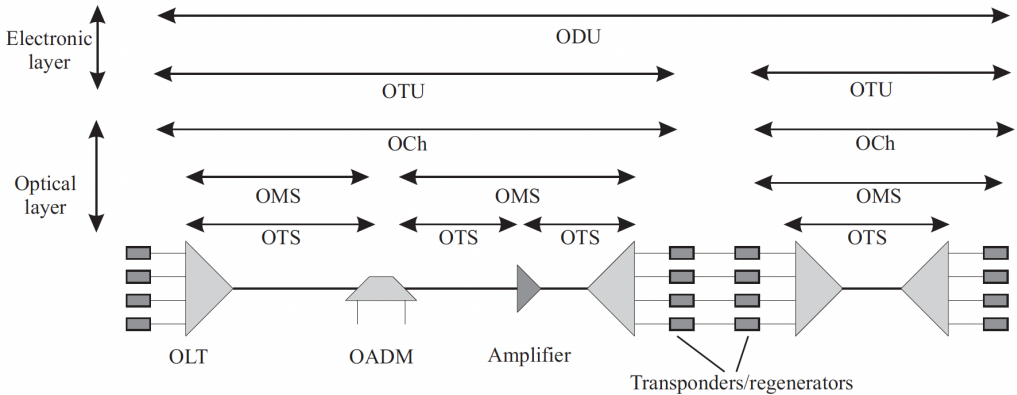
OTN Encapsulation

3 Control and Management



Layers within the Optical Layer

3 Control and Management





Performance and Fault Management

3 Control and Management

- The goal of *performance management* is to enable service providers to provide guaranteed quality of service to the users of their network
- This usually requires:
  - monitoring of the performance parameters for all the connections
  - taking any actions necessary to ensure that the desired performance goals are met
- *Fault management* involves:
  - detecting problems in the network
  - alerting the management systems appropriately through alarms
- Fault management also includes restoring service in the event of failures



BER and Optical Trace

3 Control and Management

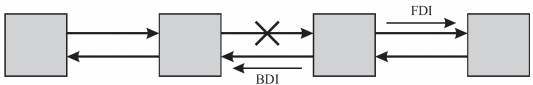
- *BER*
  - The bit error rate (BER) is the key performance attribute associated with a lightpath
  - The BER can be detected only when the signal is available in the electrical domain, typically at regenerator or transponder locations
  - Overhead inserted in OTN frames, which consists of parity check bytes, allows for BER computation
- *Optical Trace*:
  - Lightpaths pass through multiple nodes and through multiple cards within the equipment deployed at each node.
  - It is desirable to have a unique identifier associated with each lightpath
  - This identifier is called an optical path trace
  - The trace enables the management system to identify, verify, and manage the connectivity of a lightpath



Alarm Management

3 Control and Management

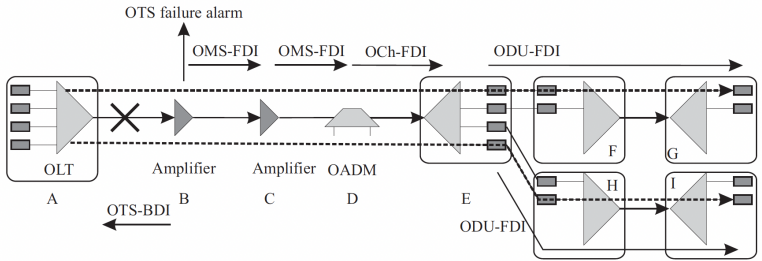
- In a network, a single failure event may cause multiple alarms to be generated
  - in a network with 32 lightpaths on a given link, each traversing through two intermediate nodes, the failure of a single link could trigger a total of 129 alarms
- Alarm management it is required to identify the root-cause alarm of the failure and suppress the redundant alarms
  - Alarm suppression is accomplished by using a set of special signals, called the forward defect indicator (FDI) and the backward defect indicator (BDI)



Alarm Management: Example

3 Control and Management

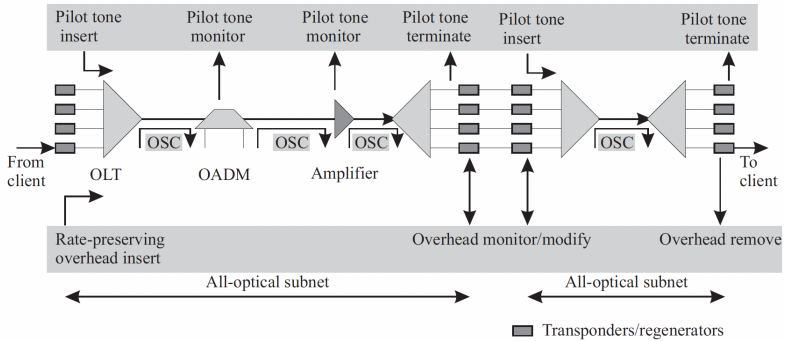
- When a link fails, the downstream node sends an FDI signal to subsequent nodes, allowing them to suppress redundant alarms.
- The same node sends a BDI signal upstream to notify the previous node of the failure.
- FDI and BDI signals operate at different sublayers of the optical layer.



Optical Layer Overhead

3 Control and Management

- Supporting the optical path trace, defect indicators, and BER measurement requires the use of some sort of overhead in the optical layer





Optical Layer Overhead

3 Control and Management

	All-Optical Subnet		End-to-End
Application	**OSC**	**Pilot Tone**	**Rate-Preserving**
Trace	OTS	OCh	OTU ODU
DIs	OTS OMS OCh	None	OTU ODU
Performance monitoring	None	Optical power	BER
Client signal compatibility	Any	Any	Any